# Imitation Learning – Sim2real

자율주행 이론 및 실습
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RLLAB
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RLLAB
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    - ROS 2 connection
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    - Input processing 2

# Sim2real

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- What we have done

1) Speed Calibration - Matching the desired speed with the actual vehicle speed through PID tuning.

2) Steering Calibration - Reflecting the maximum steering value of the actual vehicle in the simulation.

3) Min/Max vel - Modified to enable control within the desired speed range.

4) Or other parameters in f1tenth/configs/f1tenth.yaml and f1tenth/configs/dynamic.yaml

```yaml
# dynamic system
integrator: rk4
model: st

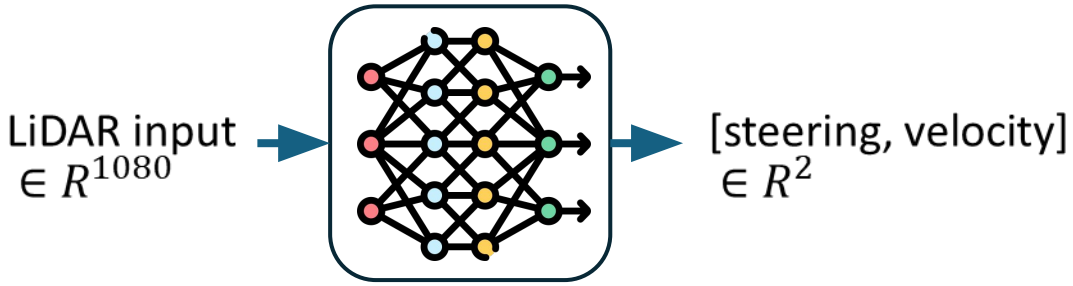
# dynmaic configs
mu: 1.0489
C_Sf: 4.718
C_Sr: 5.4562
length_f: 0.2
length_r: 0.15
h: 0.074
m: 3.74
I: 0.04712
s_min: -1.0
s_max: 1.0
sv_min: -2.65
sv_max: 2.65
v_switch: 7.319
a_max: 9.51
v_min: -8.0
v_max: 8.0
width: 0.29
length: 0.57
lidar_pos_offset: 0.155
```

□ We've made our simulation to reflect the real-world environment!

# Sim2real



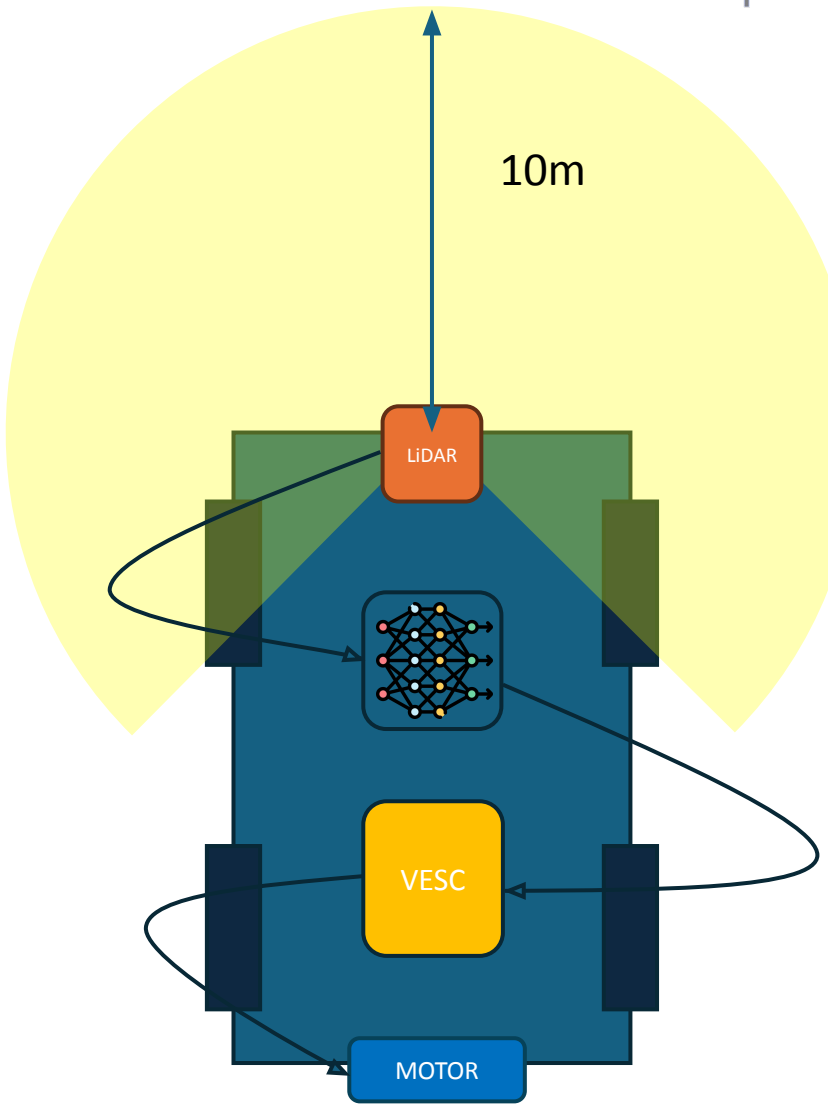
- Input processing




- Single MLP might not be enough to learn with naïve 1080 input data.

- Solutions

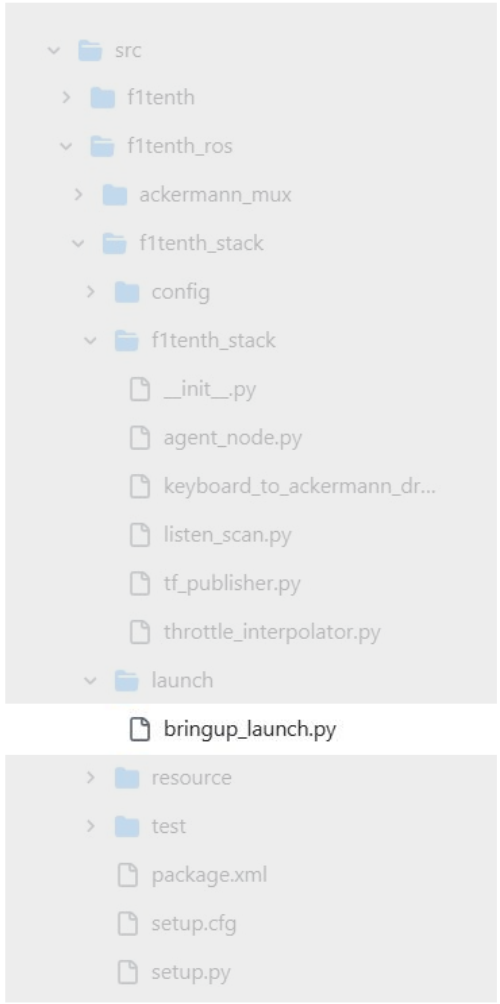
  - Input processing?
  - Other networs?
  - …



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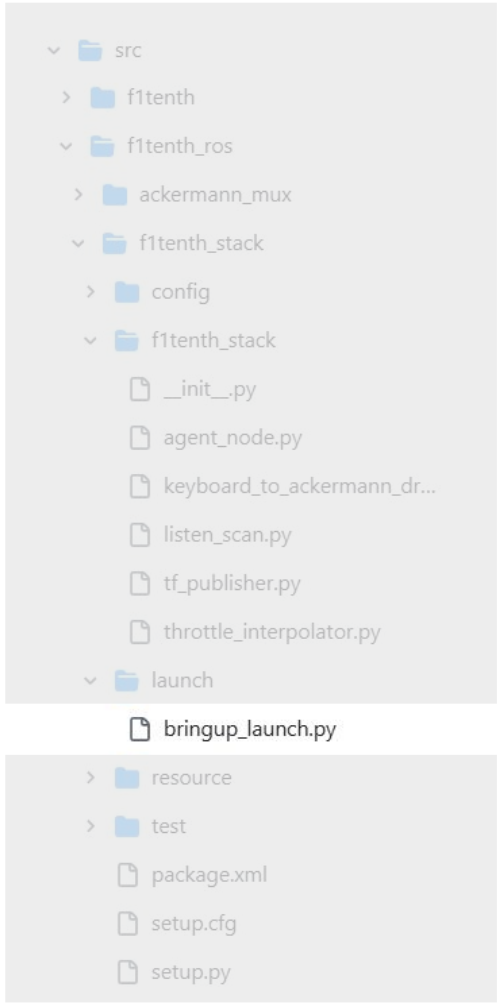
- Codes – TODO1: Connect agent_node to akermann_mux

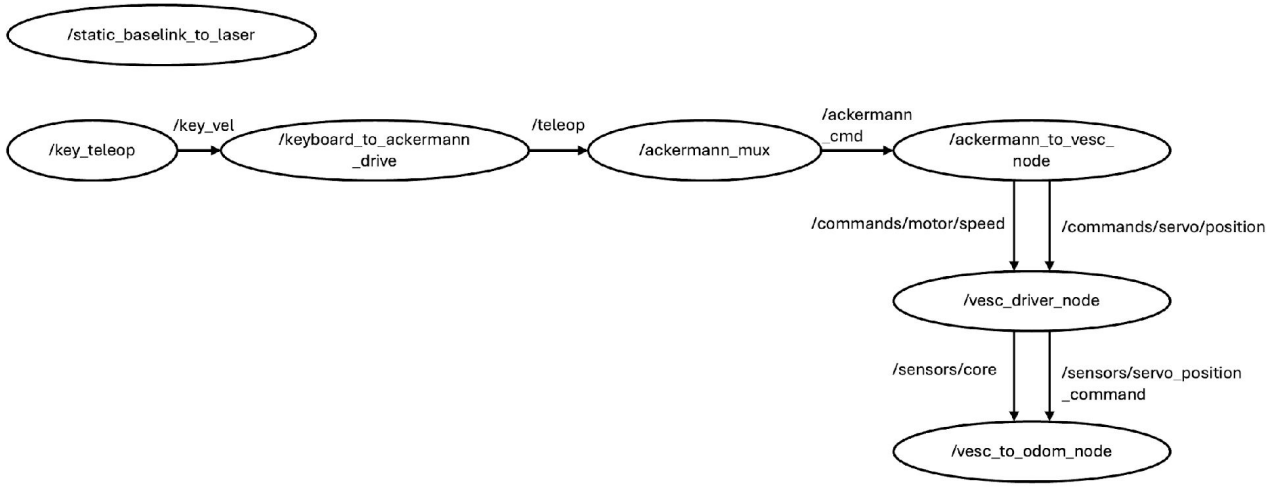


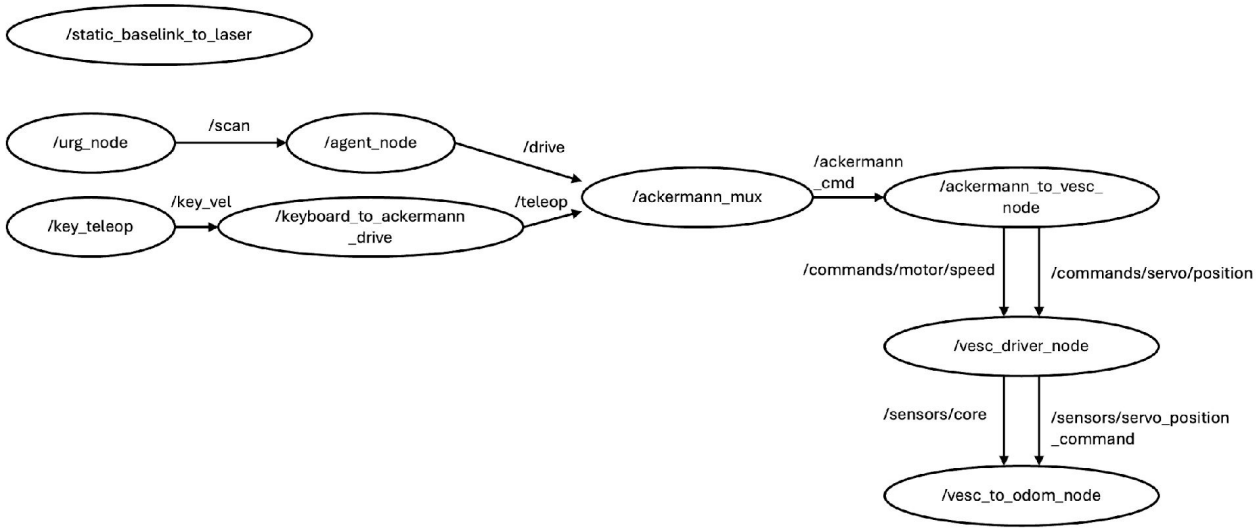

```
50          '''
51              TODO: uncomment this and add "agent_la" in LaunchDescription if you want to test your autonomous driving agent.
52          '''
53          # agent_la = DeclareLaunchArgument(
54          #     'agent_config',
55          #     default_value=agent_config,
56          #     description='Descriptions for reinforcement learning agent configs')
57
58          '''
59              TODO: replace ld
60          '''
61          ld = LaunchDescription([vesc_la, sensors_la, mux_la])
62          #ld = LaunchDescription([vesc_la, sensors_la, mux_la, agent_la])
63
64          '''
65              TODO: uncomment this if you want to test your autonomous driving agent.
66          '''
67          # agent_node = Node(
68          #     package='f1tenth_stack',
69          #     executable='agent_node',
70          #     name='agent_node',
71          #     parameters=[LaunchConfiguration('agent_config')]
72          # )
                                    ...
126         '''
127             TODO: uncomment this if you want to test your autonomous driving agent.
128         '''
129         #ld.add_action(agent_node)
130
131         return ld
```

# Sim2real

- Recall) rqt_graph



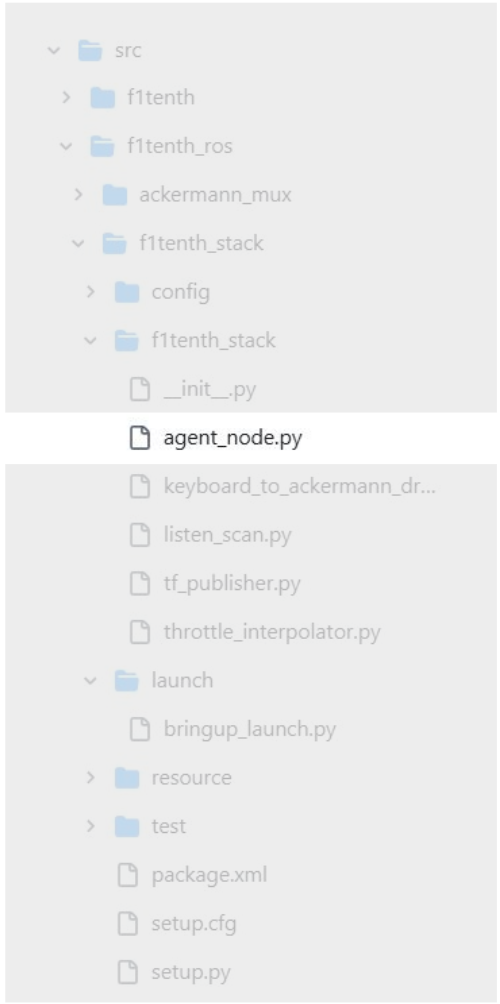



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- Codes – TODO2: Define obs dimension

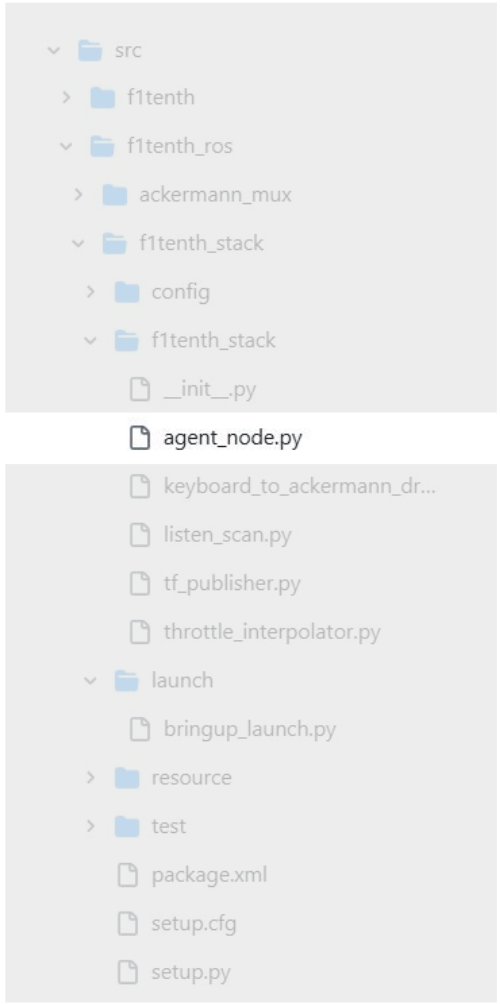



```
54
55          # observation
56          '''
57            TODO: pre-define things to process raw scan to observation.
58          '''
59          self.obs_dim =
60
61          # load parameters
```

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- Codes – TODO3: Complete getObs




```
109
110     def getObs(self, raw_scan: np.ndarray):
111         '''
112           TODO: make raw scan data to the form of observation you used to train model.
113         '''
114         return observation
115
```

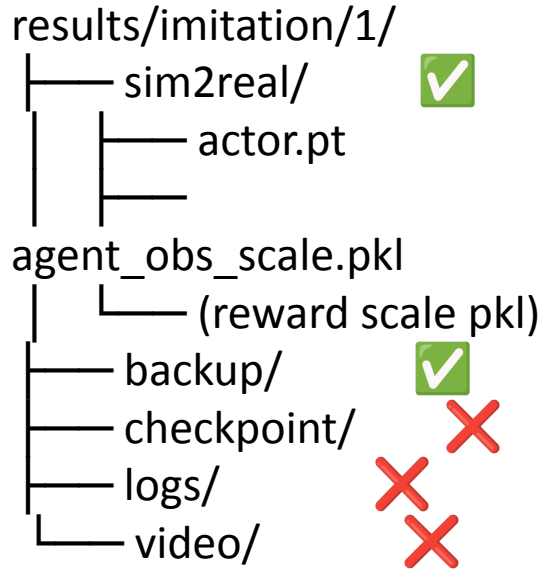
# Summary

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## TODO list

1. check if results folder placed in the right path (inside src/f1tenth folder)
2. if results folder structure looks like 'results/imitation_test/1', write 'imitation_test' to 'name' and '1' to 'algo_idx' in f1tenth_stack/config/agent.yaml
3. copy 'obs_dim' in f1tenth/configs/task/f1tenth.yaml to 'self.obs_dim' in f1tenth_stack/f1tenth_stack/agent_node.py
4. apply content of 'getObs' in f1tenth/utils/env.py to 'getObs' in f1tenth_stack/f1tenth_stack/agent_node.py
5. uncomment things related to 'agent_node' in f1tenth_stack/launch/bringup_launch.py

scp –r {경로}/f1tenth/results {team_name}@10.42.xxx.xx:./ros2_ws/src/f1tenth
Rccar 내부 src/f1tenth/algorithm 폴더 지우고
scp –r {경로}/f1tenth/algorithm {team_name}@10.42.xxx.xx:./ros2_ws/src/f1tenth



Good
Luck!